

Lingo-Blocksworld: A Dataset for Linguistically-Informed Blocksworld

Melissa Torgbi¹, Claire Benet Post², Claire Bonial³, Harish Tayyar Madabushi¹

¹University of Bath, ²University of Colorado Boulder,

³DEVCOM U.S. Army Research Laboratory

mat66@bath.ac.uk

Abstract

We introduce LINGO-BLOCKSWORLD, a linguistically-informed diagnostic benchmark for evaluating the generalisation and planning abilities of Large Language Models (LLMs) in the Blocksworld (BW) domain. Unlike prior obfuscated BW variants, our method replaces content words with morphosyntactically well-formed nonsense words, preserving the grammatical scaffolding of the domain while removing lexical familiarity. This enables us to test a psycholinguistically-motivated hypothesis and find systematic support across both reasoning and non-reasoning models, with degradation patterns that mirror syntactic bootstrapping from human language processing. Additionally, we test the ability of models to generalise planning solutions from the typical 4-block problems to 12 blocks. Overall, our results demonstrate that reasoning models are more robust where non-reasoning models collapse: non-reasoning model performance drops to near zero once problems exceed four blocks or have actions and predicates obfuscated, while reasoning models maintain stronger performance until structural and lexical difficulty are combined. Manual error analysis pinpoints further differences. Reasoning model errors are more often skipped steps or incomplete instruction following, whereas non-reasoning models fail to derive plausible plans at all. By combining psycholinguistic motivation with classical planning evaluation, LINGO-BW allows us to shift research focus from whether LLMs plan to how their planning depends on linguistic surface form.¹

1 Introduction

Large Language Models (LLMs) trained on vast amounts of data and containing billions of parameters are able to extrapolate from their training data

in ways that produce impressive performance on a number of tasks (Brown et al., 2020; Tayyar Madabushi et al., 2025; Wei et al., 2022). However, there remain tasks where LLMs fall short of human generalisation abilities (Srivastava et al., 2023; Scivetti et al., 2025), including classical planning (Sel et al., 2025). We therefore focus on planning as a controlled setting for evaluating LLM generalisation.

Planning problems involve determining the required steps to get from an initial state to a goal state (Cimatti et al., 2008) and range in complexity from very simple micro-worlds to complex real-world applications. We intentionally choose the simple planning domain of Blocksworld (BW) (Winograd, 1972; Slaney and Thiébaux, 2001) so that we can carefully vary problem difficulty in a controlled manner.

Our goal is not to evaluate LLMs as practical planners, but to use classical planning to assess their ability to generalise within a familiar planning domain beyond reliance on semantic priors. We evaluate generalisation beyond typical BW problems by incrementally altering the structural complexity and vocabulary of the problem domain. First we establish a structural baseline by varying the number of blocks. While prior work has shown that earlier models struggle with increasing structural complexity (Shojaee et al., 2025), verifying that these limits hold for newer models is a necessary calibration step for our subsequent linguistic evaluations. Next, we introduce nonsense words that maintain English morphosyntactic rules. Unlike prior obfuscated variants (Valmeekam et al., 2024a,b), this preserves the linguistic integrity of the original BW prompt for comparison. We perform ablation experiments that systematically swap out only actions, objects, predicates, and combinations thereof to investigate whether obfuscating certain linguistic cues challenges models in ways that align with psycholinguistically grounded hypothe-

¹The dataset and code to replicate our experiments are available at <https://github.com/melissatorgbi/lingoblocksworld>.

ses. This design is also motivated by limitations of existing obfuscated BW variants, such as Mystery BW and Random BW, whose lexical substitutions can either alter predicate-argument structure or fail to preserve the linguistic properties of the original domain (see [Appendix B](#)). Simultaneously, we compare non-reasoning and reasoning models to explore how different architectures respond to semantic abstraction.

Concretely, we make the following contributions:

1. We provide a novel BW evaluation dataset and generation pipeline that varies structural complexity and morphosyntactically well-formed lexical obfuscation.
2. We evaluate the zero-shot capabilities of both non-reasoning and reasoning models using our diagnostic dataset to test generalisation across increasing structural complexity and lexical novelty in the BW domain. This evaluation reveals that reasoning models are substantially better, but still degrade under combined structural and lexical difficulty. The combination of our granular dataset and manual error analysis provides novel insight into distinct failures, showing that non-reasoning models have cascading errors not present in reasoning models.
3. We introduce an increased-challenge setting that helps more precisely distinguish success on easier instances from more fragile behaviour under combined structural and lexical difficulty, beyond what previous datasets make visible.

2 Background & Related Work

LLMs have displayed impressive performance on a variety of tasks and benchmarks, but classical planning remains a challenge. Many LLMs, in fact, do not perform well on even simple planning problem domains, such as BW ([Valmeekam et al., 2023a](#)). BW is one of the original classical planning domains and remains popular for ongoing evaluation of automated planners, in particular those using the Planning Domain Definition Language (PDDL).² In BW, problems consist of multiple coloured blocks on a table. The blocks may

²BW and PDDL are leveraged more for the development of planning approaches (e.g., [Grundke et al. \(2024\)](#)), including newer approaches leveraging LLMs. For robotic applications, Signal Temporal Logic and Linear Temporal Logic are now more prevalent (e.g., [Chen et al. \(2024\)](#)).

be stacked on top of each other or directly placed on the table. Each problem has an initial configuration and a target goal state. The solution lists the actions required to get from the initial block configuration to the goal within certain constraints and leveraging a minimal set of possible actions ([Russell and Norvig, 2021](#)).

Previous research has used the BW domain to test the ability of LLMs to solve these problems. LLMs tend to fall short of human performance on these tasks in many cases ([Sel et al., 2025](#)). Large Reasoning Models (LRMs) have displayed better performance compared to non-reasoning LLMs; however, performance gains are not robust with increasing complexity and struggle to generalise well to obfuscated instances within existing BW variants ([Valmeekam et al., 2024a](#); [Shojaee et al., 2025](#)). The first obfuscated variant is Mystery Blocksworld (MYSTERY BW), in which nouns are swapped (e.g., “block” for “planet”), verbs are swapped (e.g., “stack” for “crave”), and predicates are swapped (e.g., “is clear” for “harmony”) ([Valmeekam et al., 2024b](#)). The second variant is Randomized Mystery Blocksworld (RANDOMIZED BW), which swaps out nouns, verbs, and predicates for a randomly selected string of alphanumeric characters (e.g., “arsbumagdtb748l6”).

While MYSTERY BW aims to maintain the same underlying structure as the original BW tasks, its linguistic form is neither semantically identical nor syntactically equivalent. For example, words such as “planet” or “pain” inherently carry semantic load within the embedding space of an LLM and could affect model performance. Similarly, replacing actions with randomized strings is too stringent and does not reflect real-world settings where actions are likely to remain linguistically grounded. This motivates the need for a linguistically grounded BW dataset. Table 1 showcases a comparison of the original BW domain description, the Mystery variant, the Randomized variant, and finally our own LINGO-BW variant.³

2.1 Nonce Words Obfuscation

Nonce words, which have long been used in psycholinguistic studies (e.g., [Berko \(1958\)](#)), allow us to preserve syntactic and semantic mappings while eliminating pre-existing lexical meaning. Specifically, they enable us to: (a) Duplicate the original

³For a more detailed description of issues of MYSTERY BW and RANDOMIZED BW see [Appendix B](#).

BW Variant	Sample of Domain Description
Original prompt	"Pick up a block Unstack a block from on top of another block Put down a block Stack a block on top of another block I can only pick up or unstack one block at a time. I can only pick up or unstack a block if my hand is empty."
Lingo-BW Hard (actions, objects, predicates)	"Jillerate a tegust Unbusillisk a tegust from faratived onto another tegust Astent a tegust Busillisk a tegust faratived onto another tegust I can only jillerate or unbusillisk one tegust at a time. I can only jillerate or unbusillisk a tegust if my michime is tuentsal."
Mystery BW	"Attack object Feast object from another object Succumb object Overcome object from another object To perform Attack action, the following facts need to be true: Province object, Planet object, Harmony. Once Attack action is performed the following facts will be true: Pain object."
Randomized BW	"J4gv801gnu2it0yj object_0 object_1. U64y1a9apusmslxb object_0. Kip9uw781pv62umn object_0. Wio5amhq7814n006 object_0 object_1. To perform j4gv801gnu2it0yj action, the following facts need to be true: tv30k33pzoulql6w object_1, and a46zhwykn6jvbabk object_0 Once j4gv801gnu2it0yj is performed the following facts will be true: cql4o62p1yeke3ok, tv30k33pzoulql6w object_0, and b6e9q4r60gagvden object_0 object_1"

Table 1: Parallel sections of different BW prompts with highlighted words showing matched linguistic items.

semantic and syntactic structure of BW while minimising possible contamination from memorised or pre-trained associations, and (b) Replace content words with phonotactically valid but semantically novel items, ensuring no inherited predicate-argument or affective bias. We use nonce words in our dataset to try to circumvent the issue of inherited semantic priors since nonce words theoretically have zero lexical meaning.⁴

3 Approach Overview

Having motivated the linguistic rationale for nonce-word obfuscation, we can now turn to our experimental setup that operationalises these ideas in a planning evaluation framework. We aim to target generalisation without memorisation in BW with three experiments:

1. Increasing the complexity of the problem by varying the number of blocks (§4).
2. Systematically obfuscating BW problems using nonce words (§5).
3. Combining real-world sources of complexity by increasing the number of blocks and obfuscating using nonce words (§6).

We first replicated the experiments of Valmeekam et al. (2023b) using the more recent

⁴See Appendix C for a more detailed linguistic foundation for nonce word usage.

models gpt-4.1-mini and 5-mini for comparison. We do this for the BW, MYSTERY BW and RANDOMIZED BW domains. These baseline experiments and results are discussed in Appendix F. Later experiments include the addition of the following models: o4-mini, Llama 4 Maverick, Llama 3.3 70B, Qwen2.5 32B, Qwen3 32B, DeepSeek-R1-Distill-Llama and Deepseek-R1. These models were selected to represent the range of different LLMs currently available including closed-source and open-weight models as well as reasoning and non-reasoning LLMs. Non-reasoning models in this work are autoregressive models trained primarily on the next-token prediction task followed by instruction-tuning. Reasoning models share the same foundation as non-reasoning models, but employ explicit chain-of-thought (CoT) during inference as part of their architecture. All experiments are conducted using zero-shot prompting with some additional one-shot experiments discussed in Appendix J. We verify LLM output solutions in all experiments with Fast Downward (Helmert, 2006).⁵ We note that the morphosyntactic preservation in LINGO-BW is at the prompt level, not the tokenizer level and so our methodology ensures that nonsense words appear in grammatically well-formed sentences mirroring more realistic set-ups of unfamiliar planning problems.

3.1 Hypotheses

Consistent with prior literature on LLM planning (e.g., Valmeekam et al. (2024b); Stechly et al. (2024)), we hypothesise that an increased number of blocks will lead to increased complexity, and therefore, decreased performance. However, we do not expect a linear relationship between the increasing number of blocks and decreasing performance. Given that there are a variety of planning benchmarks available online that leverage 4 blocks, fewer that explore 5 blocks, and very few or none exploring more than 5 blocks, we **hypothesise that those models relying solely on extrapolation from pre-training data (as opposed to more effective inference-time behaviour of problem-solving strategies) may show a notable drop in performance beyond 4 blocks.**

Regarding systematically obfuscating the problems using nonce words, we hypothesise that the part of speech and domain role of the nonce word

⁵Fast Downward is an algorithmic planning tool for established domains which we adapt for LINGO-BW.